数值代数练习题

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1. 设 $\boldsymbol{A}\in\mathbb{R}^{n\times n}$ 对称正定, $\boldsymbol{B}\in\mathbb{R}^{n\times n}$ 非奇异且满足: $\boldsymbol{B}^{-1}+\boldsymbol{B}^{-\mathrm{T}}-\boldsymbol{A}$ 对称正定, 求证: $\rho(\boldsymbol{I}-\boldsymbol{B}\boldsymbol{A})<1$.
解答: 注意到

$$\rho^2(\boldsymbol{I}-\boldsymbol{B}\boldsymbol{A})\leqslant\|\boldsymbol{I}-\boldsymbol{B}\boldsymbol{A}\|_2^2=\|(\boldsymbol{I}-\boldsymbol{B}\boldsymbol{A})^{\mathrm{T}}(\boldsymbol{I}-\boldsymbol{B}\boldsymbol{A})\|_2=\rho((\boldsymbol{I}-\boldsymbol{A}\boldsymbol{B}^{\mathrm{T}})(\boldsymbol{I}-\boldsymbol{B}\boldsymbol{A})),$$

故只需证: $\rho((\boldsymbol{I}-\boldsymbol{A}\boldsymbol{B}^{\mathrm{T}})(\boldsymbol{I}-\boldsymbol{B}\boldsymbol{A}))<1$. 再令 $\boldsymbol{B}=\boldsymbol{A}^{-1/2}\boldsymbol{C}\boldsymbol{A}^{-1/2}$, 其中 $\boldsymbol{C}\in\mathbb{R}^{n\times n}$ 非奇异, 则

$$\boldsymbol{B}^{-1}+\boldsymbol{B}^{-\mathrm{T}}-\boldsymbol{A}=\boldsymbol{A}^{1/2}\boldsymbol{C}^{-1}\boldsymbol{A}^{1/2}+\boldsymbol{A}^{1/2}\boldsymbol{C}^{-\mathrm{T}}\boldsymbol{A}^{1/2}-\boldsymbol{A}=\boldsymbol{A}^{1/2}(\boldsymbol{C}^{-1}+\boldsymbol{C}^{-\mathrm{T}}-\boldsymbol{I})\boldsymbol{A}^{1/2}$$

正定, 故 $\boldsymbol{C}^{-1}+\boldsymbol{C}^{-\mathrm{T}}-\boldsymbol{I}$ 也正定, 同时

$$(\boldsymbol{I}-\boldsymbol{A}\boldsymbol{B}^{\mathrm{T}})(\boldsymbol{I}-\boldsymbol{B}\boldsymbol{A})=(\boldsymbol{I}-\boldsymbol{A}\boldsymbol{A}^{-1/2}\boldsymbol{C}^{\mathrm{T}}\boldsymbol{A}^{-1/2})(\boldsymbol{I}-\boldsymbol{A}^{-1/2}\boldsymbol{C}\boldsymbol{A}^{-1/2}\boldsymbol{A})=\boldsymbol{A}^{1/2}(\boldsymbol{I}-\boldsymbol{C}^{\mathrm{T}})\boldsymbol{A}^{-1}(\boldsymbol{I}-\boldsymbol{C})\boldsymbol{A}^{1/2},$$

从而只需证: $\rho(\boldsymbol{A}^{1/2}(\boldsymbol{I}-\boldsymbol{C}^{\mathrm{T}})\boldsymbol{A}^{-1}(\boldsymbol{I}-\boldsymbol{C})\boldsymbol{A}^{1/2})<1$, 而其非零特征值又与

$$(\boldsymbol{I}-\boldsymbol{C}^{\mathrm{T}})(\boldsymbol{I}-\boldsymbol{C})\boldsymbol{A}^{1/2}\boldsymbol{A}^{-1}\boldsymbol{A}^{1/2}=(\boldsymbol{I}-\boldsymbol{C}^{\mathrm{T}})(\boldsymbol{I}-\boldsymbol{C})$$

的非零特征值相同, 进而只需证: $\rho((\boldsymbol{I}-\boldsymbol{C}^{\mathrm{T}})(\boldsymbol{I}-\boldsymbol{C}))<1$, 显然该矩阵对称半正定, 特征值均非负, 于是只需证: $\boldsymbol{I}-(\boldsymbol{I}-\boldsymbol{C}^{\mathrm{T}})(\boldsymbol{I}-\boldsymbol{C})$ 正定. 事实上,

$$\boldsymbol{I}-(\boldsymbol{I}-\boldsymbol{C}^{\mathrm{T}})(\boldsymbol{I}-\boldsymbol{C})=\boldsymbol{C}^{\mathrm{T}}+\boldsymbol{C}-\boldsymbol{C}^{\mathrm{T}}\boldsymbol{C}=\boldsymbol{C}^{\mathrm{T}}(\boldsymbol{C}^{-1}+\boldsymbol{C}^{-\mathrm{T}}-\boldsymbol{I})\boldsymbol{C},$$

又 $\boldsymbol{C}^{-1}+\boldsymbol{C}^{-\mathrm{T}}-\boldsymbol{I}$ 正定, 故 $\boldsymbol{I}-(\boldsymbol{I}-\boldsymbol{C}^{\mathrm{T}})(\boldsymbol{I}-\boldsymbol{C})$ 也正定. 因此命题得证.

2. 设实对称矩阵 $\boldsymbol{A}$ 的一个特征值为 λ , 对应的单位特征向量 $\boldsymbol{v}$, 单位向量 $\boldsymbol{x}$ 满足 $\boldsymbol{x}=\boldsymbol{v}+\mathcal{O}(\varepsilon)$, $\mu=\boldsymbol{x}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{x}$, $\boldsymbol{z}$ 满足 $(\boldsymbol{A}-\mu\boldsymbol{I})\boldsymbol{z}=\boldsymbol{x}$, $\boldsymbol{y}=\boldsymbol{z}/\|\boldsymbol{z}\|_2$, 估计 $\boldsymbol{y}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{y}-\lambda$.
解答: $\boldsymbol{A}$ 可正交对角化, 设 $\boldsymbol{A}$ 的所有特征值为 $\lambda_1=\lambda,\lambda_2,\cdots,\lambda_n$, 对应的单位正交特征向量 $\boldsymbol{v}_1=\boldsymbol{v},\boldsymbol{v}_2,\cdots,\boldsymbol{v}_n$, 并设 $\lambda_1,\cdots,\lambda_m=\lambda$, 且 $\lambda_{m+1},\cdots,\lambda_n\neq\lambda$, 其中 $1\leqslant m\leqslant n$.
设 $\boldsymbol{x}=\sum\limits_{k=1}^{n}\alpha_k\boldsymbol{v}_k$, 其中 $\alpha_k\in\mathbb{R}$, 则 $\sum\limits_{k=1}^{n}\alpha_k^2=1,(\alpha_1-1)^2+\sum\limits_{k=2}^{n}\alpha_k^2=\varepsilon^2$, 故 $\alpha_1=1-\dfrac{\varepsilon^2}{2},\sum\limits_{k=2}^{n}\alpha_k^2=\varepsilon^2$, 并且

$$\mu=\sum_{k=1}^{n}\lambda_k\alpha_k^2,\quad \boldsymbol{z}=(\boldsymbol{A}-\mu\boldsymbol{I})^{-1}\boldsymbol{x}=\sum_{k=1}^{n}\frac{\alpha_k}{\lambda_k-\mu}\boldsymbol{v}_k,\quad \|\boldsymbol{z}\|_2=\left(\sum_{k=1}^{n}\frac{\alpha_k^2}{(\lambda_k-\mu)^2}\right)^{1/2},\quad \boldsymbol{y}=\sum_{k=1}^{n}\frac{\alpha_k}{(\lambda_k-\mu)\|\boldsymbol{z}\|_2}\boldsymbol{v}_k,$$

从而

$$\lambda-\mu=\lambda\sum_{k=1}^{n}\alpha_k^2-\sum_{k=1}^{n}\lambda_k\alpha_k^2=\sum_{k=m+1}^{n}(\lambda-\lambda_k)\alpha_k^2=\mathcal{O}\left(\sum_{k=2}^{n}\alpha_k^2\right)=\mathcal{O}(\varepsilon^2),$$

因此

$$\boldsymbol{y}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{y}-\lambda=\sum_{k=1}^{n}\frac{\lambda_k\alpha_k^2}{(\lambda_k-\mu)^2\|\boldsymbol{z}\|_2^2}-\lambda$$

$$
\begin{aligned}
&= \left(\sum_{k=1}^{n} \frac{\lambda_k}{(\lambda_k-\mu)^2}\alpha_k^2\right) \Big/ \left(\sum_{k=1}^{n} \frac{1}{(\lambda_k-\mu)^2}\alpha_k^2\right) - \lambda \\
&= \left(\sum_{k=m+1}^{n} \frac{\lambda_k-\lambda}{(\lambda_k-\mu)^2}\alpha_k^2\right) \Big/ \left(\sum_{k=1}^{n} \frac{1}{(\lambda_k-\mu)^2}\alpha_k^2\right) \\
&= \left(\sum_{k=m+1}^{n} \frac{(\lambda_k-\lambda)(\lambda-\mu)^2}{(\lambda_k-\mu)^2}\alpha_k^2\right) \Big/ \left(\sum_{k=1}^{n} \frac{(\lambda-\mu)^2}{(\lambda_k-\mu)^2}\alpha_k^2\right) \\
&= \left(\sum_{k=m+1}^{n} \frac{(\lambda_k-\lambda)(\lambda-\mu)^2}{(\lambda_k-\mu)^2}\alpha_k^2\right) \Big/ \left(\sum_{k=1}^{m} \alpha_k^2 + \sum_{k=m+1}^{n} \frac{(\lambda-\mu)^2}{(\lambda_k-\mu)^2}\alpha_k^2\right) \\
&= \mathcal{O}\left((\lambda-\mu)^2 \sum_{k=m+1}^{n} \alpha_k^2\right) \Big/ \left(\alpha_1^2 + \mathcal{O}\left(\sum_{k=2}^{n} \alpha_k^2\right)\right) \\
&= \mathcal{O}\left((\lambda-\mu)^2 \sum_{k=2}^{n} \alpha_k^2\right) \Big/ \left(\alpha_1^2 + \mathcal{O}\left(\sum_{k=2}^{n} \alpha_k^2\right)\right) \\
&= \mathcal{O}((\varepsilon^2)^2 \cdot \varepsilon^2)/(1+\mathcal{O}(\varepsilon^2)) = \mathcal{O}(\varepsilon^6).
\end{aligned}
$$

3. 设 $\boldsymbol{A}$ 对称正定, Cholesky 分解为 $\boldsymbol{A} = \boldsymbol{L}\boldsymbol{D}\boldsymbol{L}^{\mathrm{T}}$, 其中 $\boldsymbol{L}$ 是单位下三角阵, D 为对角阵, 求证:

$$\kappa_p(\boldsymbol{A}) \geqslant \kappa_p(\boldsymbol{D}), \quad \forall\, p \geqslant 1; \quad \kappa_{\mathrm{F}}(\boldsymbol{A}) \geqslant \kappa_{\mathrm{F}}(\boldsymbol{D}).$$

解答: 由于 $\kappa(\boldsymbol{A}) = \|\boldsymbol{A}\|\|\boldsymbol{A}^{-1}\|$, 以及 $\boldsymbol{A}^{-1} = \boldsymbol{L}^{-\mathrm{T}}\boldsymbol{D}^{-1}\boldsymbol{L}^{-1}$, 故只需证:

$$\|\boldsymbol{A}\|_p \geqslant \|\boldsymbol{D}\|_p, \quad \forall\, p \geqslant 1; \quad \|\boldsymbol{A}\|_{\mathrm{F}} \geqslant \|\boldsymbol{D}\|_{\mathrm{F}}.$$

设

$$
\boldsymbol{L} = \begin{bmatrix} 1 & & & & \\ l_{21} & 1 & & & \\ l_{31} & l_{32} & 1 & & \\ \vdots & \vdots & \vdots & \ddots & \\ l_{n1} & l_{n2} & l_{n3} & \cdots & 1 \end{bmatrix}, \quad \boldsymbol{D} = \mathrm{diag}(d_1, d_2, \cdots, d_n), \quad d_i > 0, \quad d_m = \max_{1\leqslant i\leqslant n}\{d_i\}, \quad 1 \leqslant m \leqslant n.
$$

$\forall\, p \geqslant 1$, 都有

$$\|\boldsymbol{D}\|_p = \max_{\|\boldsymbol{x}\|_p=1} \|\boldsymbol{D}\boldsymbol{x}\|_p = \max_{|x_1|^p+\cdots+|x_n|^p=1} (|d_1x_1|^p + \cdots + |d_nx_n|^p)^{1/p} = d_m,$$

故

$$
\begin{aligned}
\|\boldsymbol{A}\|_p &= \max_{\|\boldsymbol{x}\|_p=1} \|\boldsymbol{A}\boldsymbol{x}\|_p \geqslant \|\boldsymbol{A}\boldsymbol{e}_m\|_p = \|\boldsymbol{L}\boldsymbol{D}\boldsymbol{L}^{\mathrm{T}}\boldsymbol{e}_m\|_p \\
&= \|\boldsymbol{L}[d_1l_{m1}, \cdots, d_{m-1}l_{m,m-1}, d_m, 0, \cdots, 0]^{\mathrm{T}}\|_p \\
&\geqslant |d_1l_{m1}^2 + \cdots + d_{m-1}l_{m,m-1}^2 + d_m| \geqslant d_m = \|\boldsymbol{D}\|_p.
\end{aligned}
$$

同时

$$\|\boldsymbol{D}\|_{\mathrm{F}} = \left(\sum_{i=1}^{n} d_i^2\right)^{1/2}, \quad a_{ij} = \sum_{k=1}^{\min\{i,j\}} d_k l_{ik} l_{jk}, \quad \forall\, 1 \leqslant i, j \leqslant n,$$

从而

$$\|\boldsymbol{A}\|_{\mathrm{F}} = \left(\sum_{i,j=1}^{n} a_{ij}^2\right)^{1/2} \geqslant \left(\sum_{i=1}^{n} a_{ii}^2\right)^{1/2} = \left(\sum_{i=1}^{n}\left(\sum_{k=1}^{i} d_k l_{ik}^2\right)^2\right)^{1/2} \geqslant \left(\sum_{i=1}^{n} d_i^2\right)^{1/2} = \|\boldsymbol{D}\|_{\mathrm{F}}.$$

综上, 命题得证.